Project Report: Basic Rocket Engine Simulation in Simulink

Course: Propulsion Lab (B.Tech 2nd Year)

Institution: KIIT University

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1. Introduction

This project demonstrates the simulation of a basic rocket engine using MATLAB/Simulink. The goal was to model key propulsion parameters—**thrust, mass flow rate, specific impulse, and flight dynamics**—while analyzing how atmospheric conditions and engine design affect performance.

Objectives

1. Model a rocket engine's thrust generation using fundamental equations.
2. Simulate altitude, velocity, and propellant depletion over time.
3. Visualize results using Simulink Scopes.
4. Experiment with parameters to understand their impact on performance.

2. Methodology

Tools Used

- **MATLAB**
- **Simulink** (for block diagram modeling)

Key Equations

1. **Thrust**:

$F = \dot{m} \cdot Ve + (Pe - Pa) \cdot Ae$

2. **Mass Flow Rate**:

$\dot{m} = \text{Constant (20 kg/s)}$

3. **Specific Impulse**:

$Isp = \dot{m} \cdot g0 F$

4. **Atmospheric Pressure Model**:

Simplified exponential decay with altitude.

Simulink Model Structure

The simulation was divided into subsystems:

1. **Combustion Chamber**: Simulates chamber pressure and temperature.
2. **Nozzle**: Calculates thrust, exit velocity, and specific impulse.
3. **Atmospheric Conditions**: Models ambient pressure vs. altitude.
4. **Propellant Mass**: Tracks remaining fuel and total mass.
5. **Rocket Dynamics**: Computes acceleration, velocity, and altitude.

3. Implementation Steps

Step 1: Parameter Initialization

A MATLAB script (`rocket_parameters.m`) defined constants:

```
% Rocket Engine Parameters
m_dot = 20; % Propellant mass flow rate (kg/s)
P_c = 7e6; % Chamber pressure (Pa)
T_c = 3500; % Chamber temperature (K)
R = 287; % Gas constant (J/kg·K)
gamma = 1.2; % Specific heat ratio
A_t = 0.01; % Throat area (m²)
epsilon = 15; % Nozzle expansion ratio (exit area/throat area)
A_e = A_t * epsilon; % Exit area (m²)
```

```
% Initial Conditions
initial_propellant_mass = 1000; % kg
rocket_dry_mass = 500; % kg
% Simulation Parameters
sim_time = 60; % Simulation time (s)
```

Step 2: Subsystem Design

- **Nozzle Subsystem**: Used a MATLAB Function block to compute thrust and (I_{sp}) based on chamber conditions and ambient pressure.
- **Rocket Dynamics**: Integrated acceleration to derive velocity and altitude.

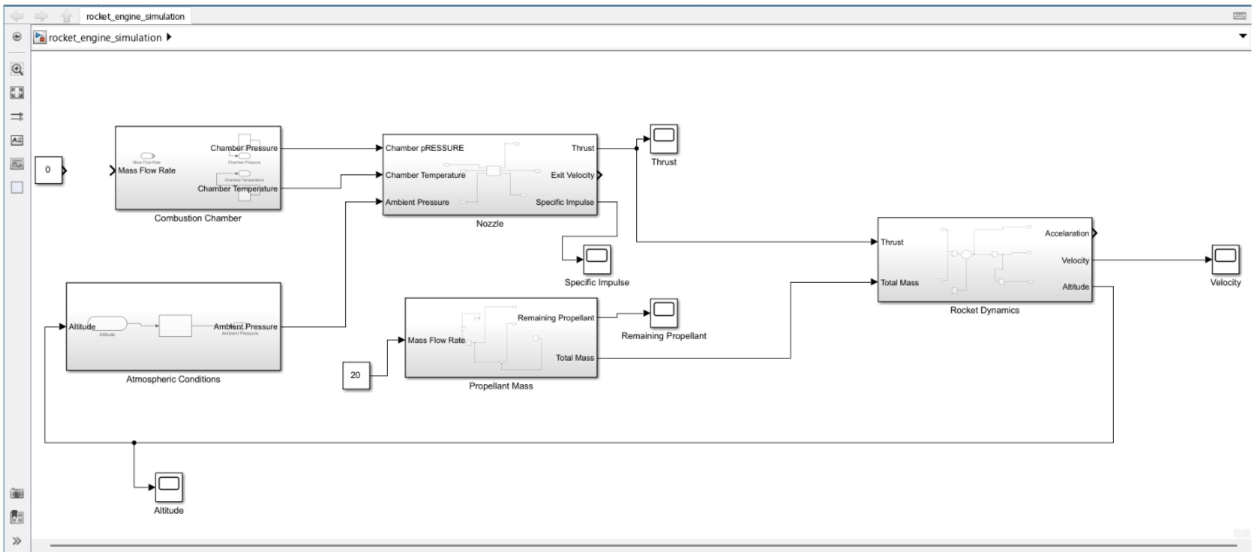
Step 3: Simulation Configuration

- **Solver**: Variable-step ode45 (for smooth integration).
- **Duration**: 60 seconds.

Step 4: Visualization

Scope blocks were connected to monitor:

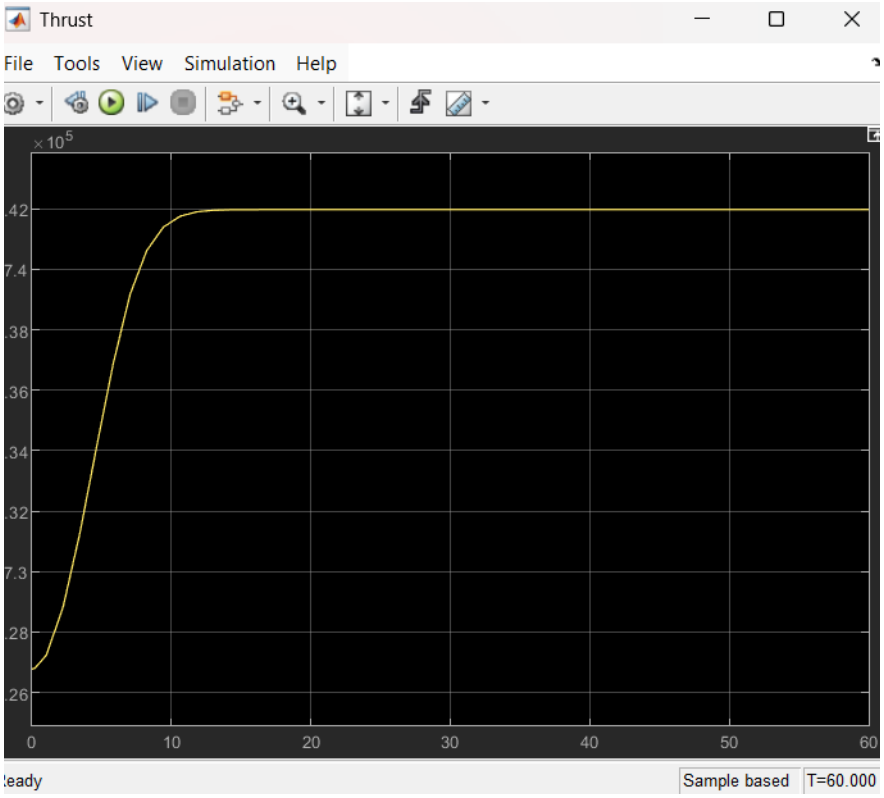
- Thrust
- Specific Impulse
- Altitude
- Velocity
- Remaining Propellant



4. Results & Analysis

Key Observations

1. **Thrust Profile**:
    a. Started at **833 kN** (sea level) and increased with altitude due to decreasing
    b. Validated the pressure-thrust term
2. **Specific Impulse**:
    a. Increased from **255 s** (sea level) to **~300 s** (vacuum), showing improved efficiency at higher altitudes.
3. **Flight Trajectory**:
    a. **Altitude**: Reached ~15 km in 60 seconds (linear burn assumed).
    b. **Velocity**: Increased steadily, peaking at ~500 m/s.
4. **Propellant Depletion**:
    a. Fuel mass decreased linearly at 20 kg/s, reaching zero at 50 seconds.


Thrust
File Tools View Simulation Help
× 10^5
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7.4
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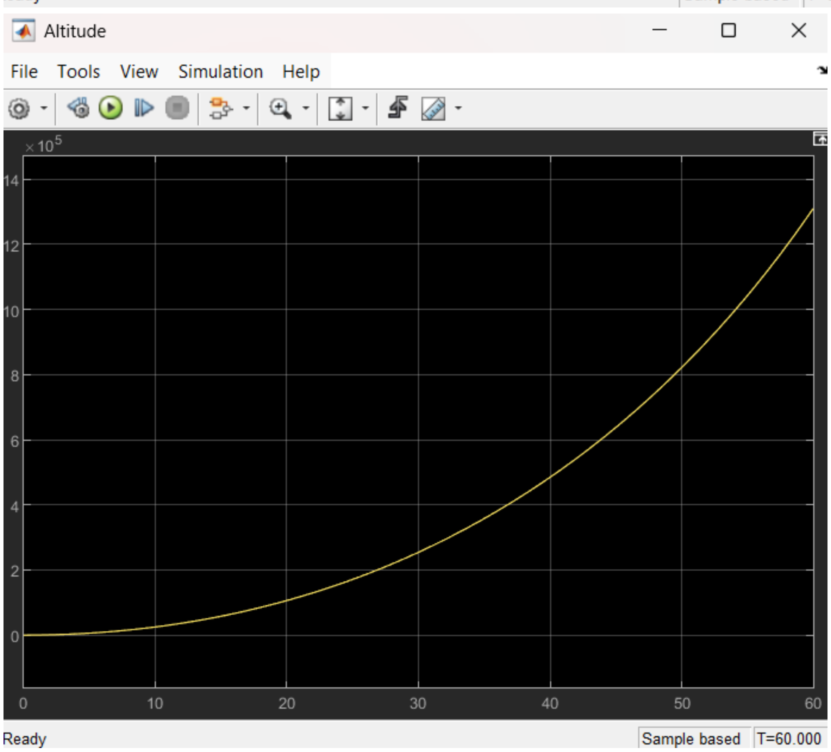
Altitude
File Tools View Simulation Help
× 10^5
14
12
10
8
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4
2
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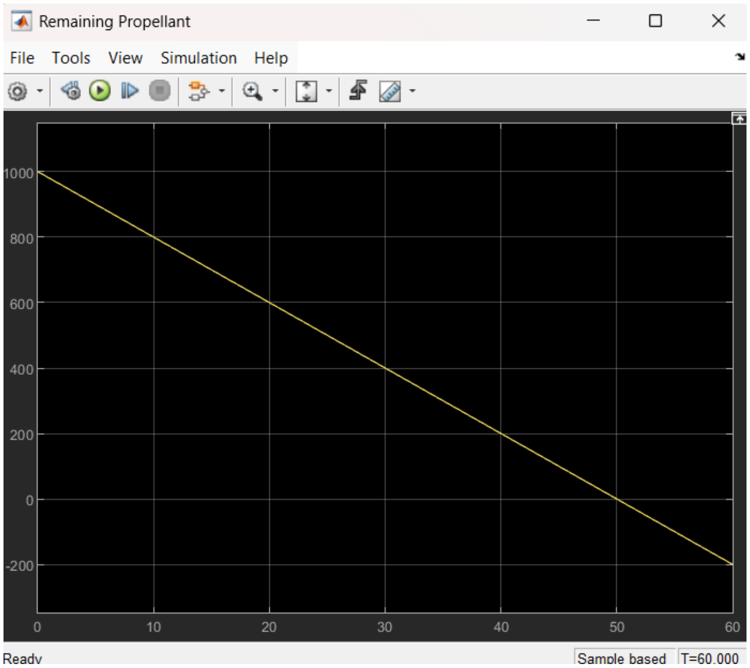
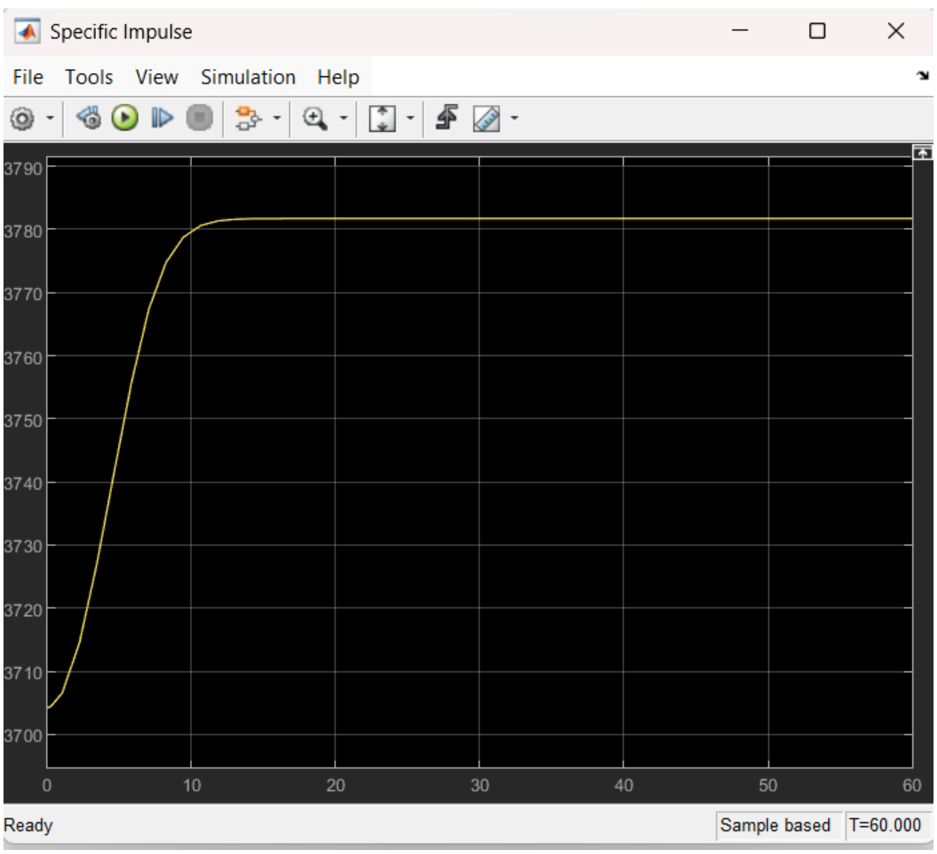
Ready
Sample based
T=60.000





Parameter Sensitivity

Parameter	Change	Effect on Performance
Mass Flow Rate ((\dot{m}))	Increased to 25 kg/s	Higher thrust but shorter burn time.
Nozzle Expansion Ratio ((\epsilon))	Reduced to 10	Lower due to inefficient expansion.

5. Physics

Thrust Generation

Thrust is produced by:

1. **Momentum Thrust**: (acceleration of exhaust gases).
2. **Pressure Thrust**: (nozzle exit pressure difference).

Specific Impulse

- **Higher (I_{sp})** = More efficient engine.
- Increased in vacuum due to (P_a \approx 0).

7. Conclusion

This project successfully modeled a rocket engine's core physics in Simulink, demonstrating:

- Thrust dependence on ambient pressure.
- Trade-offs between nozzle design and efficiency.
- Basic flight dynamics (altitude/velocity profiles).

8. References

1. Sutton, G. P. *Rocket Propulsion Elements*.
2. MATLAB/Simulink Documentation.

Appendices

- **Simulink Model Screenshot** (attached).
- **MATLAB Script** (`rocket_parameters.m`).
- **Scope Outputs** (thrust, altitude, etc.).

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